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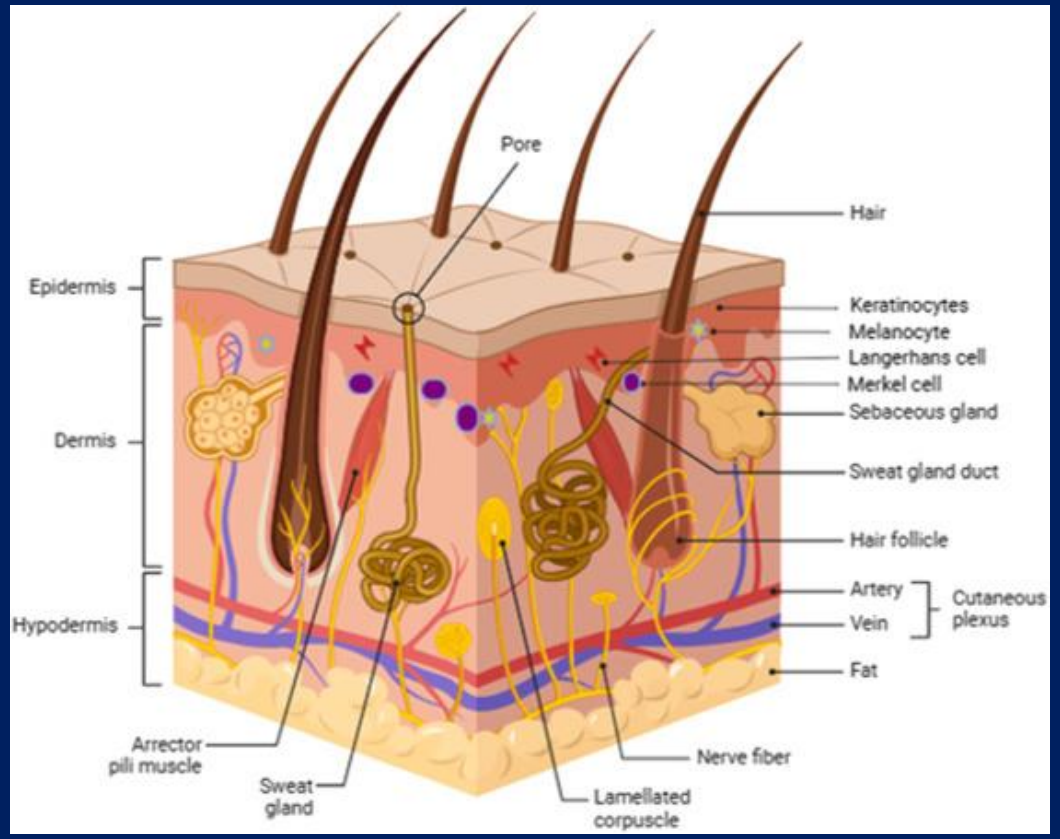
BMC

# APPROACH TO SKIN LESIONS

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# Learning Objectives

- **Knowledge:** Analyze clinical history and physical examination findings in patients presenting with dermatologic lesions.
- **Skills:** Differentiate common dermatologic conditions based on lesion morphology and anatomic distribution; formulate evidence-based diagnostic plans.

# Main Subtopics

- Dermatologic History Taking Framework
- Systematic Skin Examination Technique
- Diagnostic Investigations (Dermoscopy, Skin biopsy, Patch testing, Wood's lamp)

# Approach to skin rash



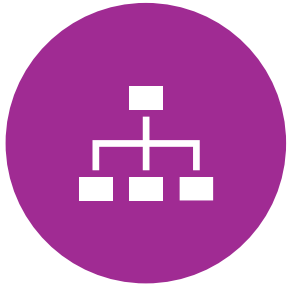
History



Examination



Investigation



Management

# Dermatologic History

- Chief complaint
- Hx of presenting Illness
- Past medical hx
- Systemic review
- Medication and allergies
- Family hx
- Social and occupational hx
- Travel and environmental exposure

# History of presenting illness

Time of onset of rash.

Is it first episode or recurrent

Time course of an individual lesiones

How long has the lesione been present?

How long did the lesion last before disappearing?

Pattern of onset and spread

How did the rash start and how has it spread?

# History of presenting illness

Location of rash onset or first lesiones

Associated symptoms

- Itchy ? , painful ? , burning ? , tingling ?

Exacerbating and relieving factors

Previous skin biopsy

Previous Treatment

# History of presenting illness

Quality of life

Does skin rash affect your sleep , activities , social life , self confidence , job ?

Assessment tool as DERMATOLOGY LIFE QUALITY INDEX ( DLQI)

# History of presenting illness

## FOR PIGMENTED OR SUSPICIOUS LESIONES FOR MALIGNANCY

- We need to ask about the evolution of the lesions
- Has it grew or changed in size ?
- Is there any changes in the shape or outline ?
- Does it changed in color ?
- New symptoms ? As bleeding , pain , itchiness

# History of presenting illness

## Examples of exacerbating factor

| | |
|---|---|
| UV exposure | Flare of lupus and rosacea |
| Temperature | Hot drink in rosacea |
| Drugs | Beta blocker induced psoriasis |
| Pregnancy | Flare of psoriasis |
| Supplements | Anabolic steroid in acne |

# History of presenting illness

Example of relieving factors :

| | |
|---|---|
| UV exposure | Sun exposure with psoriasis |
| Moisturizing | Dry skin in dermatitis |
| Drugs | Antihistamine in urticaria |

# History of presenting illness

Previous treatment

Steroid – responsive disease

Partial response causing different morphology – tinea incognito

Tx induced allergic or contact dermatitis

Hydroquinon – induced ochronosis

Systemic treatment , phototherapy and LASER

# PAST MEDICAL HISTORY and SYSTEMIC REVIEW

Related comorbidities

Cutaneous manifestation of systemic disease

As with PCOS , Diabetes mellitus

Contraindication to meds

E.g Methotrexate an liver disease

# Medication and Allergies

- Contraindication
- Drug interaction
- Drug- induced disease



# Family history

- Genetic inheritance :

- Atopic dermatitis , psoriasis , ichthyosis , melanoma

- Outbreak in family members

- Chickenpox , scabies, lice

# Occupational and social history

- Smoking

- Aggravate lupus and hidradenitis suppurativa

- Alcohol intake

- Worsening rosacea and interact with Methotrexate

- Sun exposure

- Sunburn , pigmentation , wrinkles , cancer

- Sport

- Palmar calluses in weight lifters , athlete's foot in runners

# Occupational and social history

- Work related irritant or allergic contact dermatitis
- Time relation between exposure and reaction
- improvement of rash on vacation

- Work related infection :
- Milkers nodule
- Orf

# Allergic contact dermatitis in hairdressers





Milker’s nodules ( viral infection of teats of cow – hands of farmers)



# Skin examination

- Require proper light , natural sun light is the best
- Complete skin exam is ideal to detect pattern or hidden lesiones.
- Should include mucous membrane , scalp , hair and nails
- Tools :
- - Magnifying lense , side lighting , dermatoscope ,woods lamp

# Skin examination

## Visual and tactile inspection

## Steps of morphology :

- 1ry and 2ry lesiones
- Color
- Shape
- Configuration / arrangement
- Distribution

# Skin examination

## Palpation

- Temperature ---- cellulitis , abcess
- Tenderness ---- erythema nodosum , neuroma
- Consistency --- nodular melanoma , neurofibroma , cyst
- Infiltration ---- cutaneous lymphoma , morphea
- Texture ----- pityriasis versicolor

# Skin examination

Inspection of mucosal surfaces

- Lichen planus







# Skin examination

- Inspection of nails

- Psoriasis

# Skin examination



tinea capitis



Alopecia areata



Androgenic alopecia

# Skin examination

- Hirsutism

- Source of hyperandrogenism
- Ovarian ---- PCOS
- Adrenal ---- cushing's syndrome , tumors
- Pituitary ---- hyperprolactinemia , acromegaly
- Drugs ---- anabolic steroids



# Instruments used in skin examination

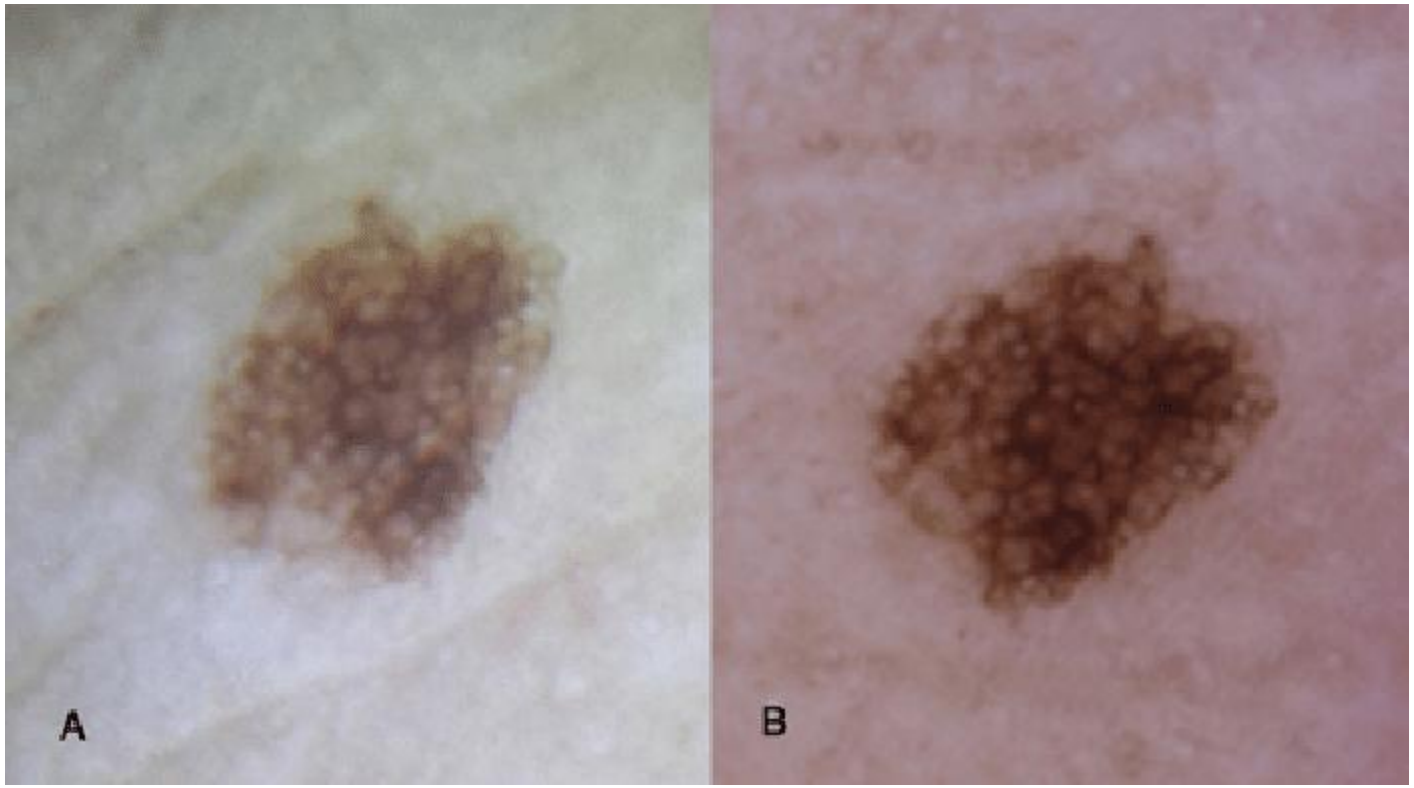




# Dermatoscopy

- Hand-held magnifying device with non- polarized light
- Skin lesions and eruption have specific dermoscopic features
- Mainly used for atypical features of suspicious pre-cancerous or cancerous lesions

# Dermatoscopy



# Wood's lamp

- Low energy UV light source
- Detect fluorescent condition or borders of some lesions
- Normal skin doesn't fluoresce

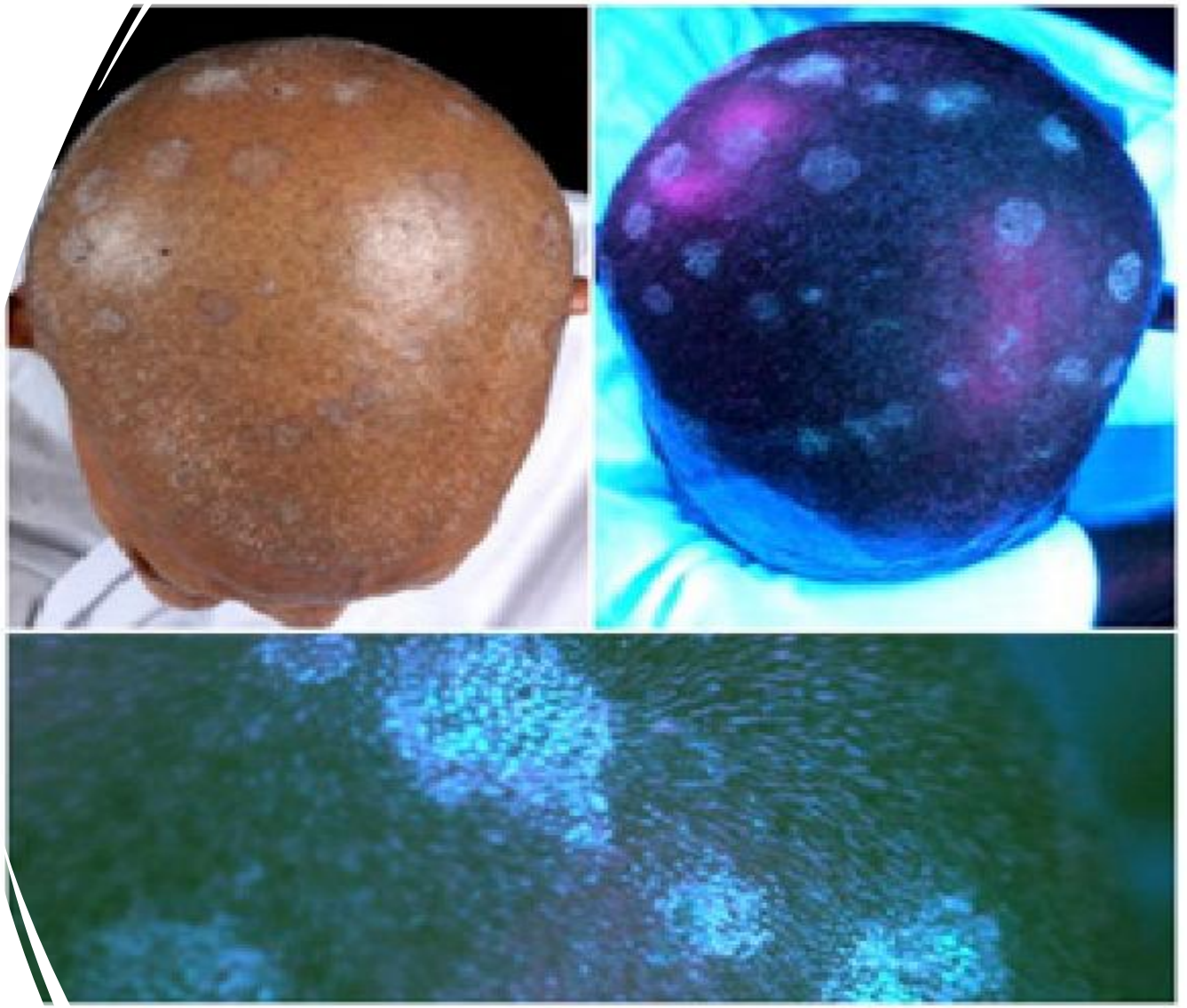
# Wood's lamp

- Example of applications:

- Red : erythrasma ( cornebacterium minutissimum )
- Yellow : pityriasis versicolor ( Malassezia furfur )
- Bright white : vitiligo
- Blue green : tinea capitis ( microsporum spp.)
- Pink : poropyrins in urine

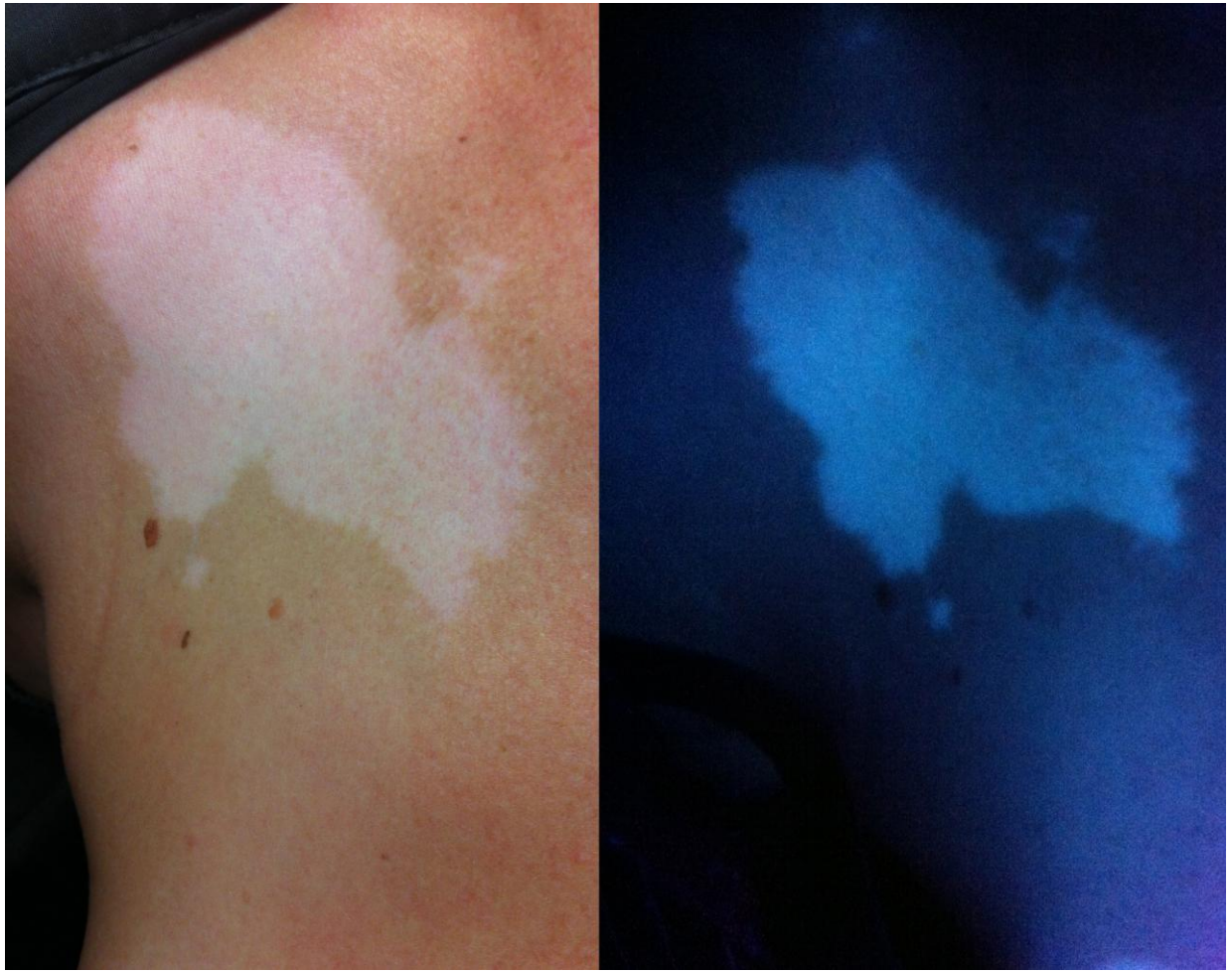
# Wood’s lamp

- Tinea capitis



# Wood’s lamp

- Vitiligo



# Dermatological signs

- Auspitz sign
- Nikolosky’s sign
- Dermatographism

# Auspitz’s sign

Elicited by removing the scales of a rash that result in pinpoint bleeding

+ve in psoriasis



# Nikolosky’s sign

- Elicited by applying lateral pressure by finger in the affected skin , result in dislodgment of upper layers of epidermis from lower layers
- In case  of blister , leads to lateral extension of blisters




- +ve in :

- Toxic epidermal necrolysis , staphylococcal scalded skin syndrome , pemphigus vulgaris

# Dermatographism

- Eliceted by stroking the skin using a tongue depressor or spatula . Within few minutes , a wheals and flare reaction develops

- +ve in urticaria



# Investigations

- Skin biopsy is very useful test in aiding diagnosis

- Do you know some indication ?

# Indication for skin biopsy

- Suspicious skin growth to rule out or confirm malignancy .
- Diagnosis of blistering disorders ( bullous pemphigoid )
- Diagnosis of some inflammatory disorders
- Confirmation of tissue infections.

# Skin biopsy

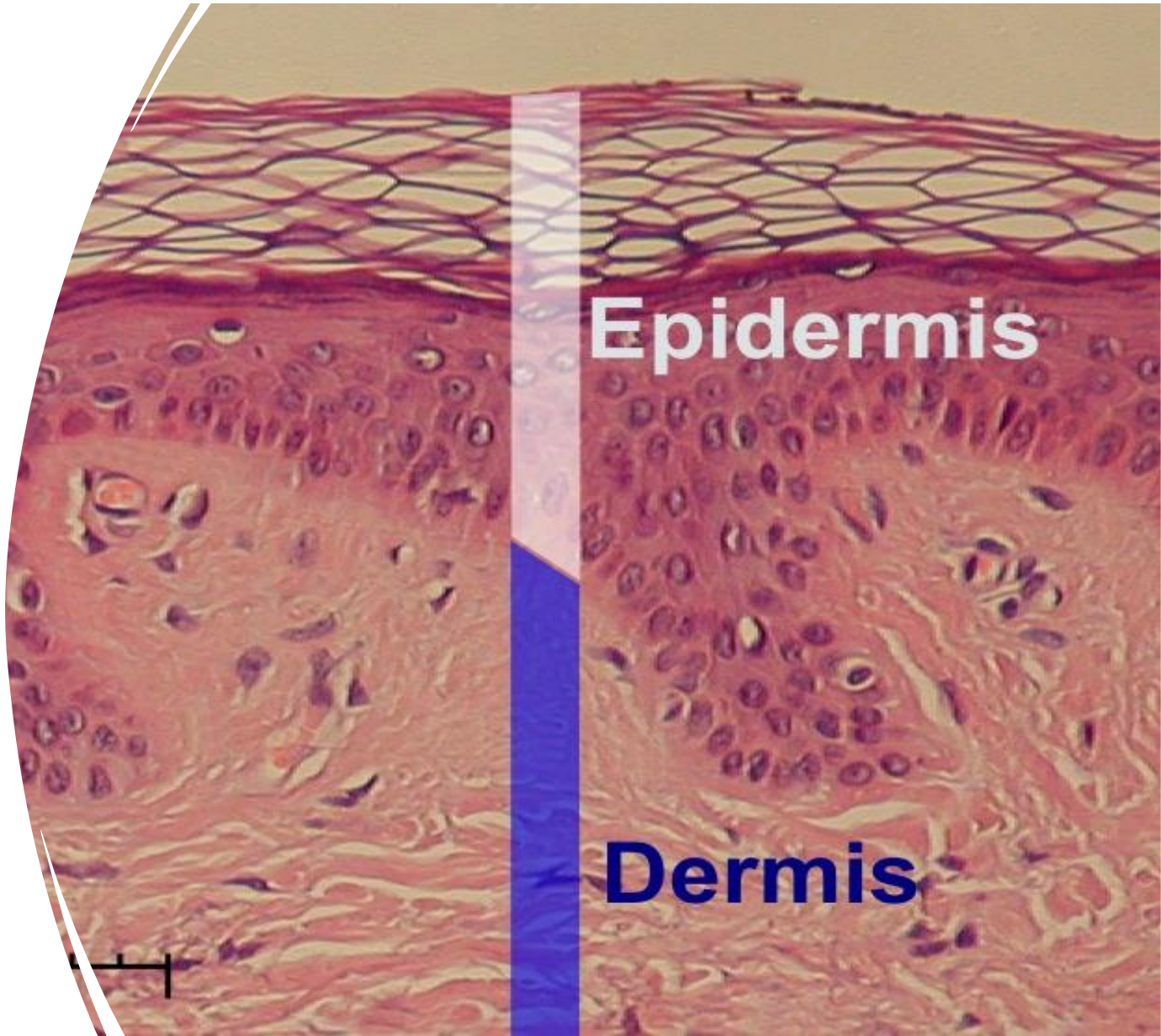
- Different type of biopsies , depending on the type and depth of the lesions or eruptions

- 3 main tests of skin specimens available

1- histopathological exam :

✓ tissue fixed in formalin

✓ H&E STAIN Used

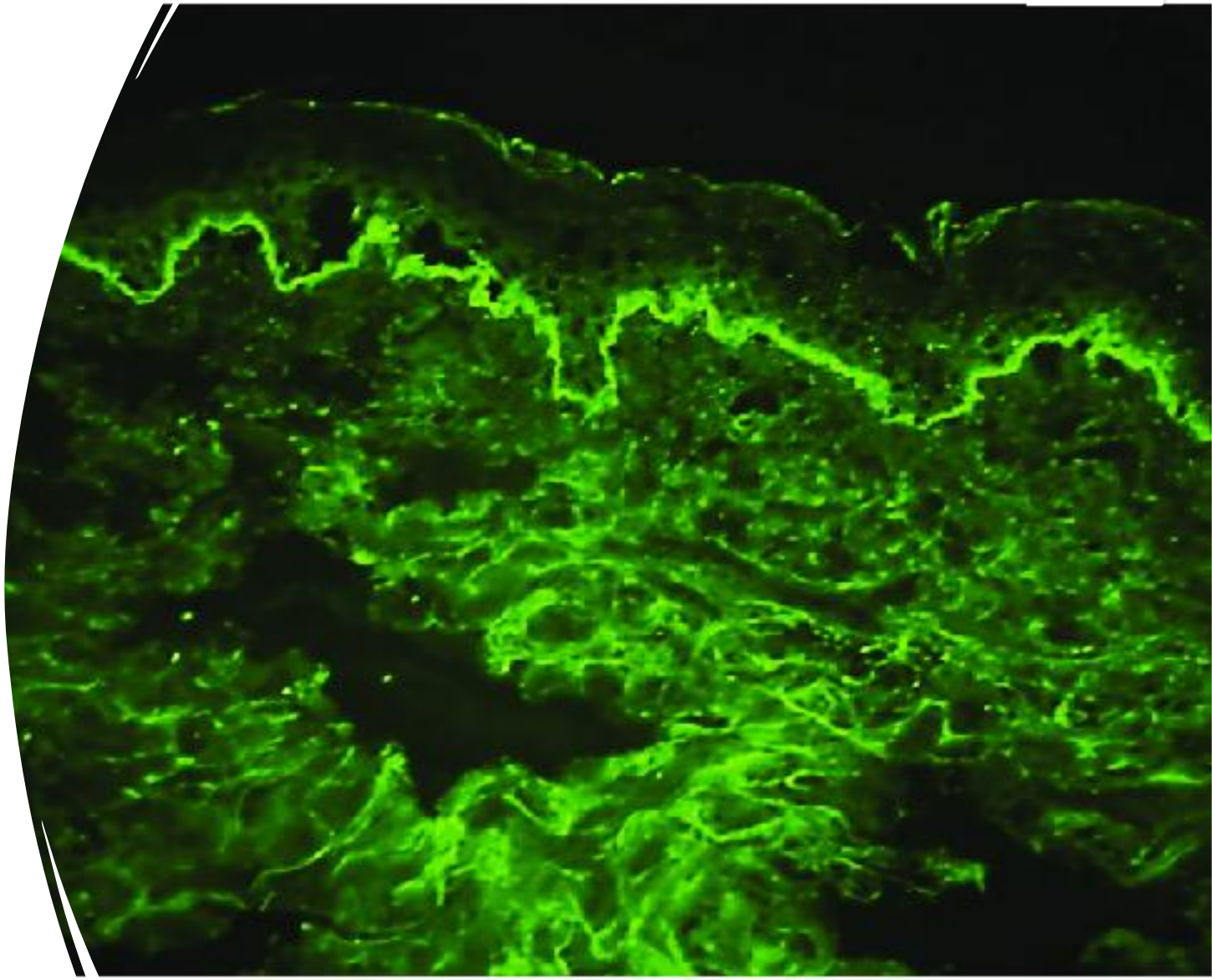


# Skin biopsy

2- direct immunofluorescence :

Tissue submitted in michel’s solution

Use specific AB that are chemicallt conjugated to flurecent dye



# Skin biopsy

3- for tissue culture

In case of infection

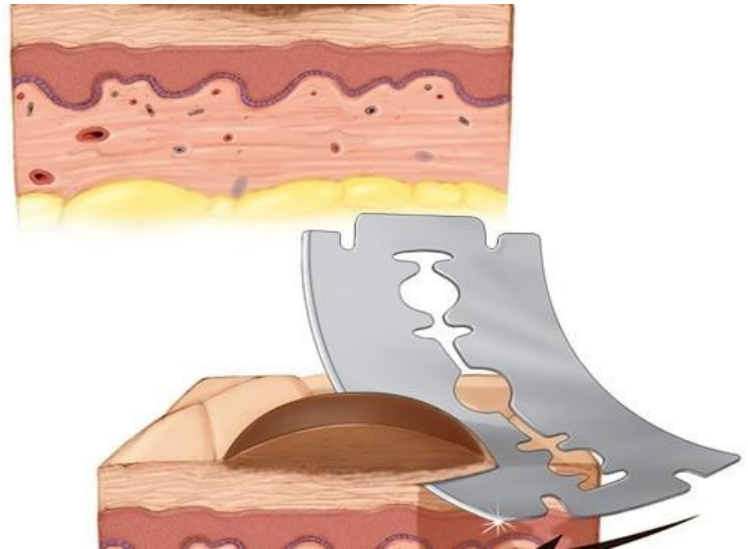
Tissue be sent in normal saline .

# Types of skin biopsy

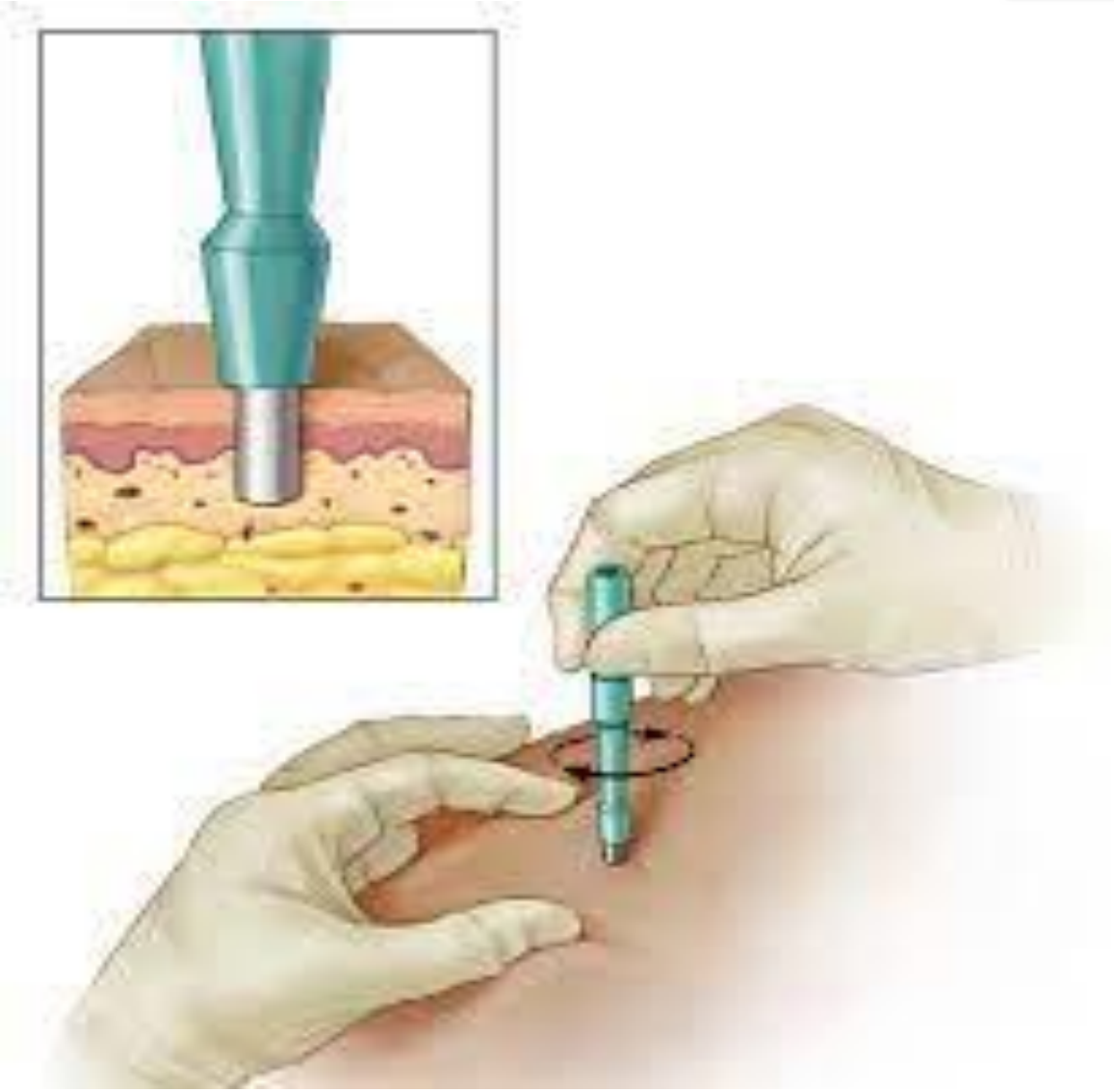
- Shave biopsy :

- Using razor ( sample contain epidermis and portion of dermis)





# Types of skin biopsy



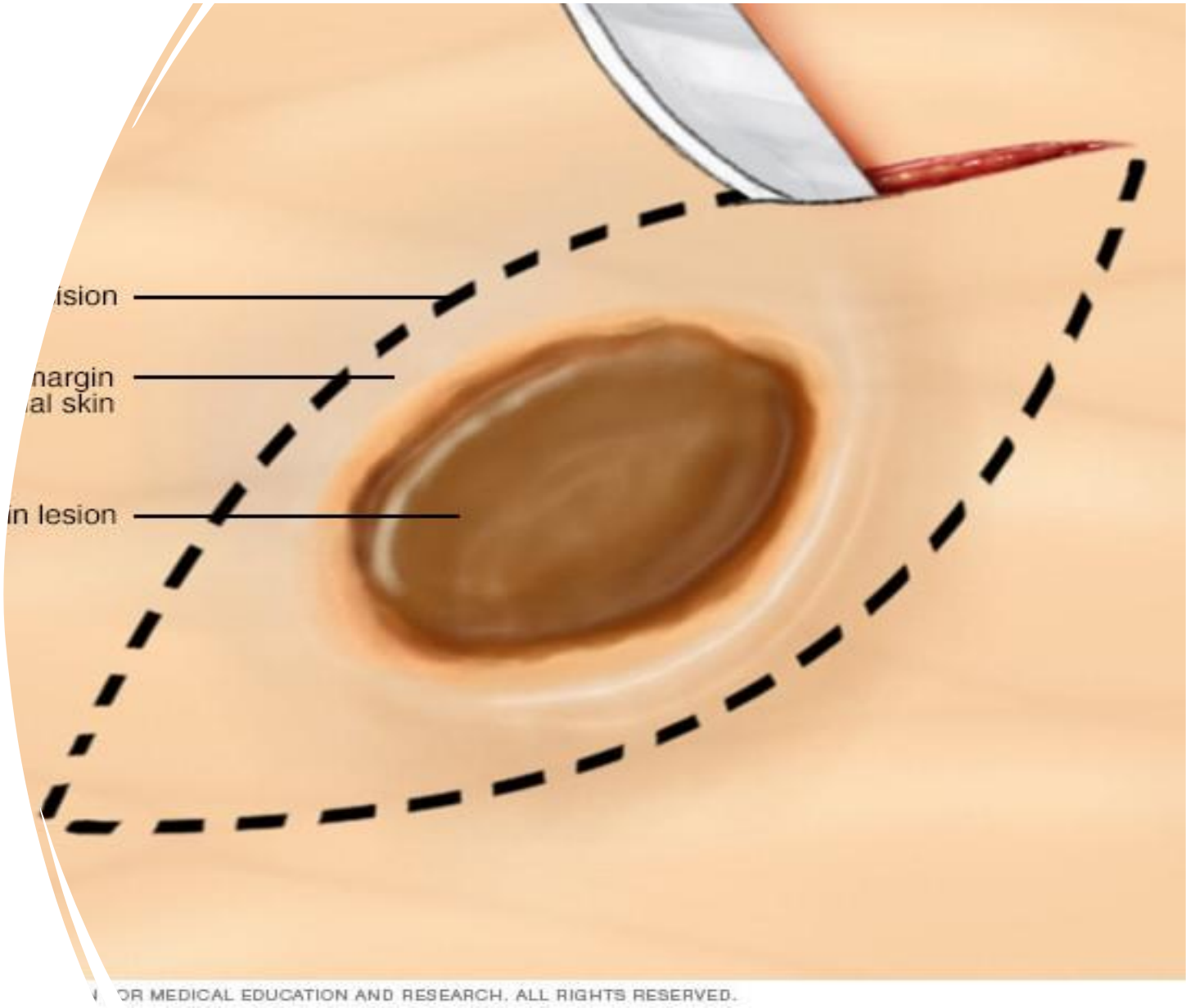
- Punch biopsy :

- Using circular punch tools ( sample contain all 3 layers of skin : epidermis , dermis and superficial fat)

# Types of skin biopsy

- Excisional biopsy :

- Using scalpel ( sample contain an entire lump or big deep block of abnormal skin)




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# Blood workup

- Screening test for underlying systemic cause ( e.g. itching , hair loss)
- Screening for associated systemic disease ( e.g. lichen planus and hepatitis c )
- Hormonal studies for acne / hirsutism ( e.g. PCOS)
- Monitoring side effect of medication (e.g. isotretinoin or methotrexate)

# Microbiological test

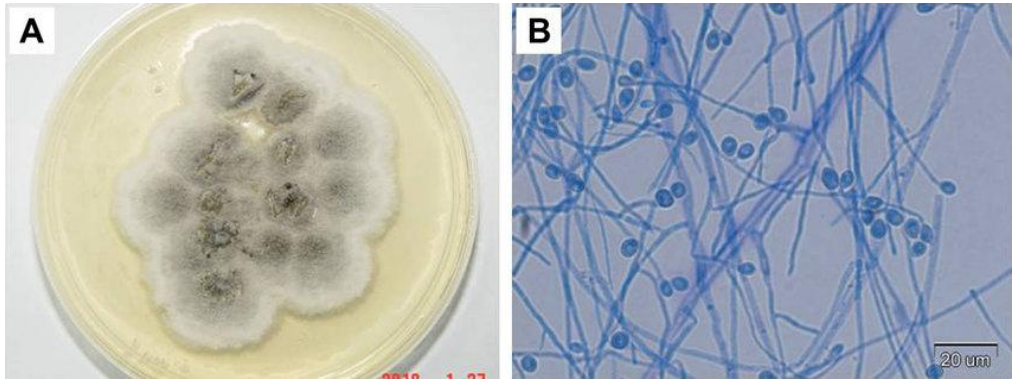
- Swab for microscopy or culture and sensitivity
- Swab or scraping of vesicles base for viral test
- Skin scraping , plucked hair and nail for fungal test
- Bedside direct microscopy for scabies and fungus

# Fungi

- Samples : skin scrapping , nail clipping , plucked hair

- Test :

- Potassium hydroxide for microscopic exam
- Fungal culture
- Skin biopsy with fungal stain

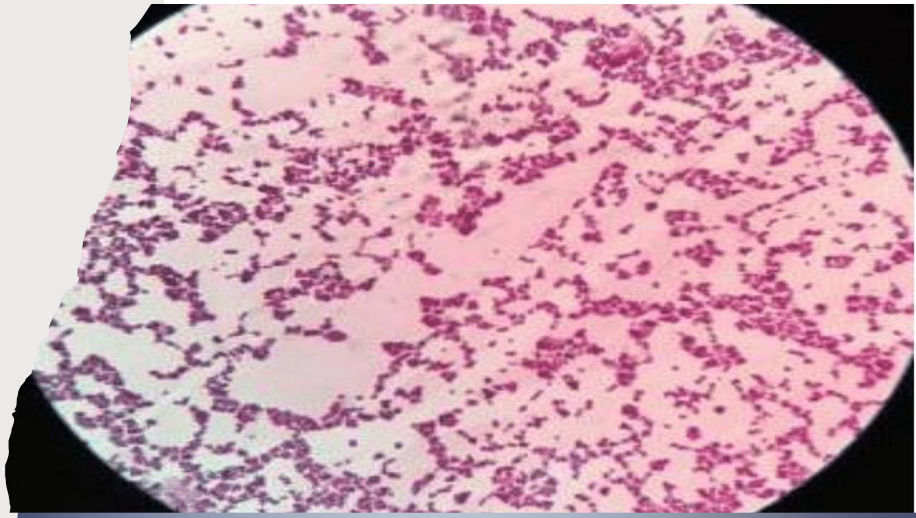


# Bacteria and mycobacteria

- Samples : swab from vesicles , pustules , erosions and ulcers

- Test :

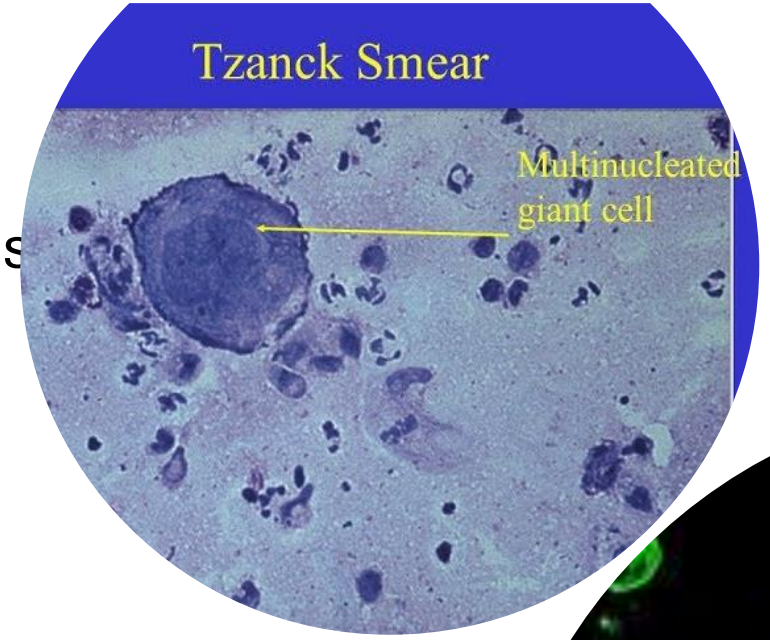
- Staining and microscopy
- Bacterial or mycobacterial culture
- Skin biopsy for tissue culture





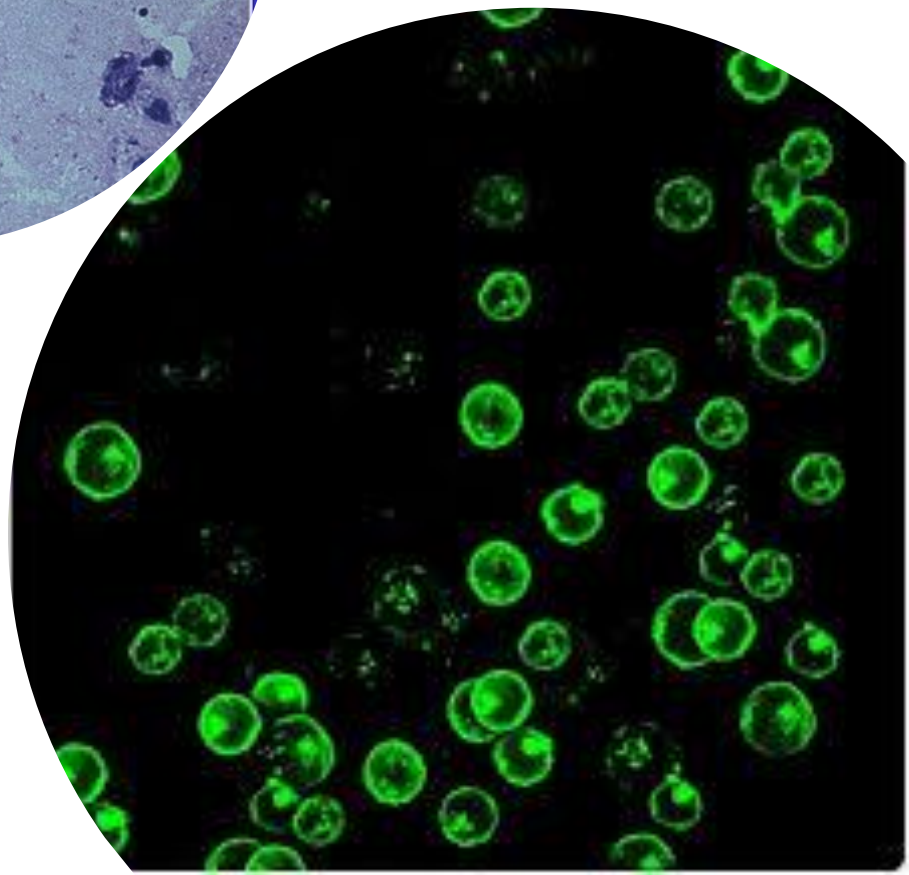
# Viruses

- Samples : swab or scrape from bas of vesicles , erosions and ulcer




- Test :

- Tzank smear
- PCR
- Direct fluorescent antibody
- Viral culture
- Skin biopsy for culture or histopathology



# Summary

- Detailed Clinical History:

- Focus on symptom onset, progression, triggers, and comprehensive medical/medication background.

- Systematic Physical Exam & Morphology:

- Conduct a thorough skin assessment to accurately identify primary lesions (macules, papules, plaques, nodules).

- Targeted Diagnostic Tools:

- Utilize bedside aids like dermoscopy and definitive tests like biopsies to confirm the diagnosis.

# Case scenario

- 42-year-old obese man with type 2 diabetes presents with a 3-month history of a well-demarcated, reddish-brown, slightly scaly patch in his left inguinal fold.
- Bedside Diagnostic Tool: Examination under Wood's Lamp in a darkened room exhibits a striking bright coral-red fluorescence.



# Case scenario

- What is the diagnosis and the underlying cause of this specific fluorescence?
- A) Tinea cruris (Trichophyton rubrum)
- B) Erythrasma (Corynebacterium minutissimum)
- C) Intertriginous Candidiasis (Candida albicans)
- D) Inverse Psoriasis

# References


- Dermatology 3rd ed jean L. Bolognia
- Fitzpatrick's dermatology

# Thank you